WILDML

Artificial Intelligence, Deep Learning, and NLP

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Recurrent Neural Network Tutorial, Part 4 – Implementing a GRU/LSTM RNN with Python and Theano

The code for this post is on Github. This is part 4, the last part of the Recurrent Neural Network Tutorial. The previous parts are:

- Recurrent Neural Networks Tutorial, Part 1 – Introduction to RNNs
- Recurrent Neural Networks Tutorial, Part 2 – Implementing a RNN with Python, Numpy and Theano
- Recurrent Neural Networks Tutorial, Part 3 – Backpropagation Through Time and Vanishing Gradients

In this post we'll learn about LSTM (Long Short Term Memory) networks and GRUs (Gated Recurrent Units). LSTMs were first proposed in 1997 by Sepp Hochreiter and Jürgen Schmidhuber, and are among the most widely used models in Deep Learning for NLP today. GRUs, first used in 2014, are a simpler variant of LSTMs that share many of the same properties. Let's start by looking at LSTMs, and then we'll see how GRUs are different.

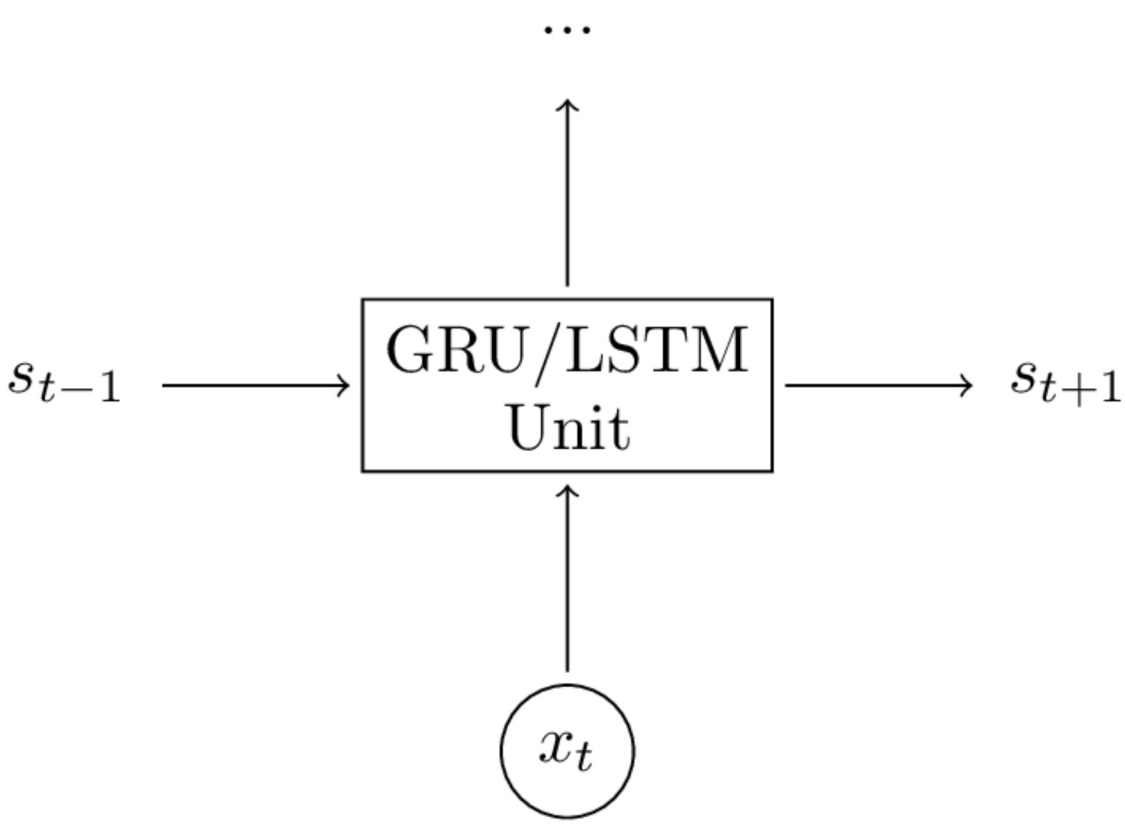
LSTM networks

In part 3 we looked at how the vanishing gradient problem prevents standard RNNs from learning long-term dependencies. LSTMs were designed to combat vanishing gradients through a *gating* mechanism. To understand what this means, let's look at how a LSTM calculates a hidden state s_t (I'm using $\circ$ to mean elementwise

multiplication):

$$
\begin{aligned}
i &= \sigma(x_t U^i + s_{t-1} W^i) \\
f &= \sigma(x_t U^f + s_{t-1} W^f) \\
o &= \sigma(x_t U^o + s_{t-1} W^o) \\
g &= \ tanh(x_t U^g + s_{t-1} W^g) \\
c_t &= c_{t-1} \circ f + g \circ i \\
s_t &= \tanh(c_t) \circ o
\end{aligned}
$$

These equations look quite complicated, but actually it's not that hard. First, notice that a LSTM layer is just another way to compute a hidden state. Previously, we computed the hidden state as $s_t = \tanh(Ux_t + Ws_{t-1})$. The inputs to this unit were x_t, the current input at step t, and s_{t-1}, the previous hidden state. The output was a new hidden state s_t. A LSTM unit does the exact same thing, just in a different way! **This is key to understanding the big picture.** You can essentially treat LSTM (and GRU) units as a black boxes. Given the current input and previous hidden state, they compute the next hidden state in some way.



With that in mind let's try to get an intuition for *how* a LSTM unit computes the hidden state. Chris Olah has an excellent post that